Seed, Envelope, Detail: Coarse-to-Fine Diffusion Attention for Training-Free Segmentation

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Abstract

Text-to-image diffusion models localize words in cross-attention (CA) and group pixels in self-attention (SA), so the images they generate can be segmented without training. We show that how these signals are combined across spatial scales *matters more than which graph operator is applied to them. Our training-free pipeline builds a class-word CA seed from the coarse-dominated modules, weighted by their agreement with SA. It then propagates the seed coarse to fine through SA at 16, 32 and 64 pixels. A one-sided cap lets each finer level redistribute evidence only within the mass its coarse parent assigns, and a fixed DenseCRF aligns the result with image edges. Each stage has a measurable role. Coarse SA supplies a high-recall object envelope that is robust to clutter, and fine SA restores detail. The cap removes 13% of background mass but only 0.5% of object mass, whereas exact mass conservation forces a wrong parent's mass onto its children and costs up to five points. DenseCRF gains six times more on capped maps in cluttered scenes. Selecting every choice on five class-stratified folds of VOC-Sim, the pipeline reaches 0.829 out-of-fold mIoU and transfers frozen to COCO-Cap at 0.696, against 0.704 and 0.582 for class-word OVAM. An oracle analysis shows that the hierarchy's gain is cross-image calibration of the propagated maps, which also explains its one systematic failure, objects that fill the frame.*

1. Introduction

A diffusion model that draws "a photograph of a sheep" must decide, at every denoising step, which image positions belong to the word *sheep* and which positions belong together. The first decision is visible in cross-attention (CA) between image queries and text keys; the second in self-attention (SA) between image positions. Reading both out yields a segmentation mask of the generated image without any training [6, 10], which makes generated images a source of free pixel labels.

Turning attention into masks is nevertheless a design problem with many degrees of freedom. Stable Diffusion 1.5 exposes 31 denoising calls, CA and SA at four spatial scales, sixteen modules of each type and eight heads per module. Existing pipelines average CA over modules, propagate it once with fine SA and refine the result [6]. Recent work adds re-weighting, graph sharpening or random walks on top of that recipe [4, 5, 7, 8]. We tested many such operators under a strict protocol, and most did not help. What did help was respecting the different roles that attention plays at different *scales* (Fig. 1):

- **Seed.** CA at 8 and 16 pixels marks the queried object reliably but coarsely; 64-pixel CA adds detail but is noisy, and 32-pixel CA hurts. Weighting modules by how well their spatial structure agrees with SA puts 84% of the weight on the coarse modules (Sec. 5.1).
- **Envelope.** Propagating the seed through coarse 16-pixel SA covers the whole object, even in cluttered scenes, but blurs its boundary; fine 64-pixel SA is precise but misses parts (Sec. 5.2).
- **Detail within the envelope.** Passing from coarse to fine, we let each finer level move evidence only *within* the mass its parent assigns, and never add mass beyond it (a one-sided cap). Enforcing exact conservation instead makes children inherit wrong parents' mass and fails (Sec. 5.3).
- **Boundary.** A fixed DenseCRF aligns the mask with image edges, and gains most once the cap has removed far-away mass (Sec. 5.4).

The resulting pipeline uses no training, no optimized tokens and no tuned graph parameters. On VOC-Sim it reaches 0.829 out-of-fold mIoU, and on COCO-Cap it reaches 0.696 in frozen transfer, against 0.704 and 0.582 for OVAM with class words [6]. Every choice is made on VOC-Sim folds, with COCO-Cap used only after freezing.

Our contributions are: (i) a coarse-to-fine SA propagation with a one-sided mass cap, motivated and isolated by controlled ablations against mass-conserving and uncapped alternatives; (ii) a stage-by-stage account, with quantitative and qualitative evidence, of what the CA seed, coarse

SA, fine SA and DenseCRF each contribute; (iii) an oracle threshold analysis showing that the hierarchy improves cross-image calibration rather than within-image ranking, which explains its main failure case; and (iv) a fold-safe selection protocol, together with the large negative record it produced.

2. Related work

Attention maps as segmentations. DAAM aggregates diffusion CA over layers and steps to attribute image regions to prompt words [10]. OVAM adds a separate attribution prompt, so that any word can be queried, fuses CA with 64-pixel SA, refines with DenseCRF, and optionally optimizes tokens on annotated images [6]. It also released VOC-Sim and COCO-Cap, the benchmarks we use. We keep its class-word setting and its DenseCRF, and change how CA is aggregated and how SA is applied across scales.

Selecting and weighting attention. GoCA weights CA layers by the agreement between their features and a self-attention structure [7]; our A3 seed adapts this idea to the full denoising trajectory and a class-word query. NERVE weights heads by semantic output and propagates with a stopped random walk [5]. We tested adaptations of both in our setting (supplementary material): the layer weighting improved the final mask, while the head weighting and the random walk did not.

Propagation over self-attention graphs. SA is a row-stochastic affinity, so repeated application spreads a seed and eventually leaks across boundaries. SPARK sharpens the affinity by expansion, inflation and pruning [4]. M2N2 treats SA as a Markov chain and uses arrival times [2]. PEARL studies attention geometry for training-free segmentation [9]. Multi-resolution use of SA appears in FA-Seg, which fuses refinements from several resolutions with fixed weights [1], and in SeeDiff, which re-extracts seeds between scales [8]. Our hierarchy instead passes one map from coarse to fine. Its key ingredient, the one-sided cap, makes the coarse level an upper bound on what the finer level may claim. We compare it directly with exact (two-sided) mass conservation.

Boundary refinement. DenseCRF combines a unary prior with appearance-dependent pairwise potentials [3]. Attention pipelines apply it with fixed parameters [6]; we do the same and analyze when it helps.

3. Method

We segment the region of a generated image that belongs to a class word w (*e.g.* "sheep"). All attention is recorded while the image is generated; nothing is trained.

Attention in the denoiser. At denoising call t, an attention module m with $n_m{=}s_m^2$ image positions forms per-head queries Q and keys K. Cross-attention uses text keys from an attribution prompt that contains w [6], and self-attention uses image keys:

$$a_{t,m} = \tfrac{1}{8}\textstyle\sum_h \mathrm{softmax}\big(Q_h K_h^\top/\sqrt{d}\big)_{[:,\,w]} \in \mathbb{R}^{n_m}, \quad (1)$$

$$A^s_{t,h} = \mathrm{softmax}\big(Q_h K_h^\top/\sqrt{d}\big) \in \mathbb{R}^{s^2\times s^2}.$$

Here $[:,w]$ averages the columns of all tokenizer pieces of w (softmax over the 77 padded text positions, conditional branch). $A^s_{t,h}$ is row-stochastic, so $A^s_{t,h}x$ replaces each value of a map x by an affinity-weighted average. We write $\bar{A}^s_t = \frac{1}{8}\sum_h A^s_{t,h}$ for the head mean, U for bilinear $2\times$ upsampling and D for area downsampling.

3.1. Semantic seed

CA modules exist at 8, 16, 32 and 64 pixels. We use the eleven modules at 8, 16 and 64 pixels, $\mathcal{M}$ (Sec. 5.1), and all 31 calls, and weight them per call:

$$c = \mathrm{norm}\left(\tfrac{1}{31}\textstyle\sum_{t=0}^{30}\sum_{m\in\mathcal{M}} w_{t,m}\, U_{64}(a_{t,m})\right),$$

$$w_{t,m} = \big[\langle \tilde{S}_{t,m}, P_t\rangle\big]_+ \big/ \textstyle\sum_{m'\in\mathcal{M}}\big[\langle \tilde{S}_{t,m'}, P_t\rangle\big]_+, \quad (2)$$

where norm is min–max normalization and U_{64} resizes to 64×64. $\tilde{S}_{t,m}$ is the head-mean SA affinity of the transformer block containing CA module m, resized to the reference grid. $P_t = \mathrm{softmax}(FF^\top/\sqrt{d})$ is a pseudo-affinity of the decoder's 32-pixel query features F at the same call. These "A3" weights, a GoCA-style adaptation [7], favour CA modules whose spatial grouping matches the network's own. We also tested GoCA's transposed orientation; it changes the seed by less than 10^{-3} (Pearson $r{=}0.99999$) and the results not at all (supplementary material).

3.2. Coarse-to-fine propagation with a one-sided cap

A single 64-pixel SA module propagates the seed with fine detail but also leaks along fine textures (Sec. 5.2). We therefore start at the coarsest SA level that still separates objects, and pass the map up one resolution at a time. For each SA call $t\in\mathcal{T}=\{10,\dots,20\}$, with $c_{16}=D(c)$,

$$r_t = \bar{A}^{16}_t c_{16}, \qquad q_t = \bar{A}^{32}_t\, U(r_t), \qquad z_t = \mathrm{cap}(r_t, q_t), \quad (3)$$

and the final map is $y = \frac{1}{|\mathcal{T}|}\sum_t \bar{A}^{64}_t\, U(z_t)$. The cap compares every 2×2 group g of 32-pixel children with its 16-pixel parent:

$$\mathrm{cap}(r,q)_k = q_k \cdot \min\Big(1,\ \frac{r_{g(k)}}{\frac{1}{4}\sum_{k'\in g(k)} q_{k'} + \epsilon}\Big), \quad (4)$$

with $\epsilon = 10^{-12}$. Bilinear upsampling preserves local means, so $4r_g$ is the parent's mass at child resolution. Equation (4) is then the generalized-KL projection of q onto the

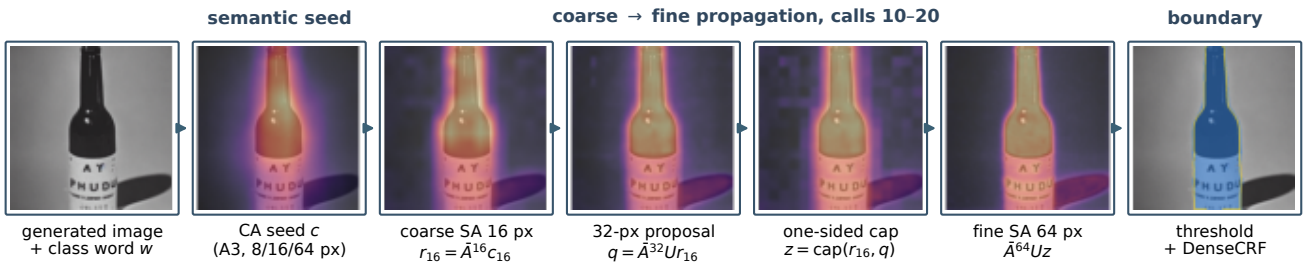


Figure 1. **Seed, envelope, detail, boundary.** Real intermediate maps for one VOC-Sim image. A class-word CA seed c (Sec. 3.1) is propagated by coarse 16-pixel SA, which spreads it over the whole bottle. The 32-pixel SA proposal q is then capped so that no 2×2 group exceeds its coarse parent's mass, which trims the background. Fine 64-pixel SA restores the outline, and DenseCRF aligns the thresholded mask with image edges (yellow: annotation). Maps are averaged over SA calls 10–20 and all heads.

upper constraints $\sum_{k\in g} z_k \le 4r_g$: groups within budget are untouched, and groups over budget are scaled down uniformly. The clipped mass is discarded, not redistributed. Two alternatives frame this choice. Without a cap, the 32-pixel level may place evidence wherever its affinities lead, including into background that the coarse level had excluded. With *exact conservation*, $\sum_{k\in g} z_k = 4r_g$, the children must also fill every parent's budget, including the budget of a coarse parent that was wrong. Section 5.3 shows that only the one-sided version helps. We cap the $16 \to 32$ transition only, and leave $32 \to 64$ uncapped so that fine SA can refine boundaries.

3.3. Mask and boundary refinement

The map is min–max normalized, bilinearly upsampled to image size and thresholded at a global τ chosen on VOC-Sim training folds (τ=0.55). DenseCRF [3] then refines the binary mask b by minimizing

$$E(x) = \sum_i \psi_u(x_i \mid b_i) + \sum_{i<j} [x_i \neq x_j]\, k(i,j), \quad (5)$$

$$k(i,j) = w_1 \exp\Big(- \frac{\|p_i - p_j\|^2}{2\theta_\alpha^2} - \frac{\|I_i - I_j\|^2}{2\theta_\beta^2} \Big) + w_2 \exp\Big(- \frac{\|p_i - p_j\|^2}{2\theta_\gamma^2} \Big),$$

with positions p, colours I and OVAM's fixed parameters (w_1=10, θ_α=80, θ_β=13, w_2=3, θ_γ=3, five mean-field iterations) [6]. We call this step R2. No ground truth enters extraction, propagation or refinement.

4. Experimental setup

Benchmarks and metric. We use the two benchmarks released with OVAM [6]: VOC-Sim, with 490 annotated images of the 20 VOC classes from prompts "a photograph of a [class]", and COCO-Cap, with 707 annotated images generated from COCO captions, often cluttered and sometimes without the class name in the caption. We regenerate each image with Stable Diffusion 1.5 from its released prompt and seed (30 steps, guidance 7.5) to record attention. The metric is the mean over 20 classes of the class IoU, pooled over all images of the class.

Fold-safe selection. VOC-Sim is the only selection set. Its images are split once into five class-stratified folds. A design choice and its threshold are chosen on four folds and scored on the fifth. Pooling the five held-out folds gives the out-of-fold (OOF) estimate, and *forced OOF* scores a named variant in every fold. A change was promoted when it gained at least 0.010 mIoU on the training folds (or at least 0.005 with 12 of 20 classes improved and no class losing more than 0.05), won at least four of five folds, and improved the pooled OOF score. COCO-Cap is scored once, with the method and threshold frozen on VOC-Sim. Every comparison in Secs. 5.1–5.5 is between cells of one run on the same generated images. Paired intervals are 95% bootstrap intervals over images (20,000 resamples).

Selection status. The CA scales, the SA module, the two-pass baseline, DenseCRF and the SA call window were promoted by this rule. Two components were adopted after the rule had retained a simpler reference. The A3 weights won 4/5 folds and transferred, but the multi-candidate screen kept uniform weights. The capped hierarchy won 5/5 folds against both references, but one class (sheep, Sec. 5.5) lost more than the 0.05 guard. We label both as post-registration decisions and report the registered outcomes in full in the supplementary material, together with 45 registered families of alternatives, most of which failed.

5. What each stage contributes

Figure 2 shows the pipeline on images chosen by a fixed rule, and Table 1 measures each decision against the previous one in the same run. The CA seed marks the class but is incomplete. Coarse SA spreads it over the object, fine SA restores the outline, and DenseCRF aligns it with edges.

no 32-px CA	SA 64 px	2 passes	DenseCRF	calls 10–20	A3 weights	C2F + cap	VOC	Δ	COCO
							.665		–
✓							.691	+.026	–
✓	✓						.789	+.033[a]	–
✓	✓	✓					.800	+.012	.625
✓	✓	✓	✓				.813	+.014	.637
✓	✓	✓	✓	✓			.817	+.004	.644
✓	✓	✓	✓	✓	✓		.823	+.006	.662
✓	✓	✓	✓	✓	✓	✓	**.829**	+.006	**.696**

Table 1. **The pipeline, one decision at a time.** VOC: OOF mIoU; Δ: gain over the previous row *measured in the same run* (same generated images); COCO-Cap: VOC-frozen transfer. The first three rows come from early one-axis screens at a fixed threshold 0.45 (CA only, then one SA pass); [a]against one pass through all sixteen SA modules (.756). Later rows select thresholds inside the training folds. Runs differ by at most 0.0003 on a shared reference.

CA resolutions (px) 8	16	32	64	CA seed only, VOC-Sim mIoU@.45	AP
✓				.563	.846
	✓			.649	.906
		✓		.300	.822
			✓	.173	.560
✓	✓	✓	✓	.666	.922
✓	✓			.693	.917
✓	✓		✓	**.696**	.913

Module weights $w_{t,m}$	CA only	+ propagation, R2	COCO
uniform	.714	.817	.644
A3 (SA agreement, Eq. 2)	.705	**.823**	**.662**

Table 2. **Building the semantic seed.** *Top:* CA-only mIoU at threshold 0.45 and average precision for resolution subsets (all calls and heads; full VOC-Sim, descriptive). Each scale alone is weak and 64-px CA is nearly unusable alone, but 8, 16 and 64 px are complementary; dropping the 32-px modules raises out-of-fold (OOF) mIoU from 0.665 to 0.691 (4/5 folds). *Bottom:* module weighting, forced OOF in one run (MID-P2 propagation of Sec. 4). A3 lowers the CA-only score yet improves the final mask in 4/5 folds and on COCO-Cap: it favours modules whose spatial structure SA can propagate.

This section explains each step in turn.

5.1. Seed: which CA scales, and how to weight them

CA is the only source of class identity, so the seed sets an upper bound on everything downstream. Table 2 (top) shows that no single CA scale suffices. The 16-pixel modules are the strongest alone, 8-pixel modules localize coarsely, and 64-pixel CA alone is nearly unusable (0.17 mIoU) because it fires on texture (see the supplementary seed gallery). The 32-pixel modules hurt in every combination: removing them raises OOF mIoU from 0.665 to 0.691 in 4/5 folds (paired $+0.022$, CI $[+0.015, +0.029]$). Restricting CA heads or calls did not survive once crossed with this choice.

Uniform averaging still lets the noisy 64-pixel modules contribute five of eleven votes. The A3 weights of Eq. (2) learn nothing, yet they shift 84% of the weight to the 8- and 16-pixel modules, most strongly at early, noisy calls (supplementary material). Interestingly, A3 *lowers* the CA-only score (0.714→0.705) but raises the final mask from 0.817 to 0.823 (4/5 folds) and COCO-Cap from 0.644 to 0.662 (Table 2, bottom). A seed is good if SA can propagate it, not if it is the best mask on its own. This is the first sign that coarse structure is the right base.

5.2. Envelope: why start from coarse self-attention

SA at different scales groups pixels differently. Figure 3 (left, middle) compares propagating the same seed through 16-pixel SA only, 64-pixel SA only, and from 16 to 64 pixels. Coarse SA keeps recall high as the threshold rises (0.73 at τ=0.70 on VOC-Sim, against 0.61 for fine SA), because it assigns whole regions, not textures, to the object. Its precision is low, however: blocks along the boundary are all-or-nothing, and alone it scores only 0.723 OOF. Fine SA is precise but loses object parts at high thresholds. Going from coarse to fine keeps most of the coarse recall at nearly fine precision. On cluttered COCO-Cap, where fine SA leaks into look-alike textures, it already beats fine-only propagation (0.687 vs. 0.660 frozen). On VOC-Sim, however, it falls short (0.802 vs. 0.810 OOF). The coarse envelope is useful, but on its own it lets the finer levels claim too much. The cap addresses this.

5.3. Detail within the envelope: cap, not conservation

How strongly should the coarse level constrain the next? Table 3 answers this in one run. The plain hierarchy (no constraint) loses to fine-only propagation on VOC-Sim. Forcing exact mass conservation at both transitions is far worse (0.759 raw, −5 points). Conservation at $16 \rightarrow 32$ alone, or conservation followed by fine completion, stays below fine-only after R2. The one-sided cap is the only variant that beats both fine-only propagation and the plain hierarchy in every fold: 0.830 after R2, $+0.012$ over the plain hierarchy (CI $[+0.006, +0.013]$), with 0.697 on COCO-Cap. A second cap at $32 \rightarrow 64$ adds nothing, which confirms that fine SA should refine boundaries freely. Re-adding completion on top of the cap re-introduces false positives and erases the COCO-Cap gain.

Why does the one-sided form work? The coarse parent is a reliable *upper bound* on where the object can be, but not a reliable lower bound. The cap binds in 58% of 2×2 groups yet removes only 7.6% of the proposal mass. As

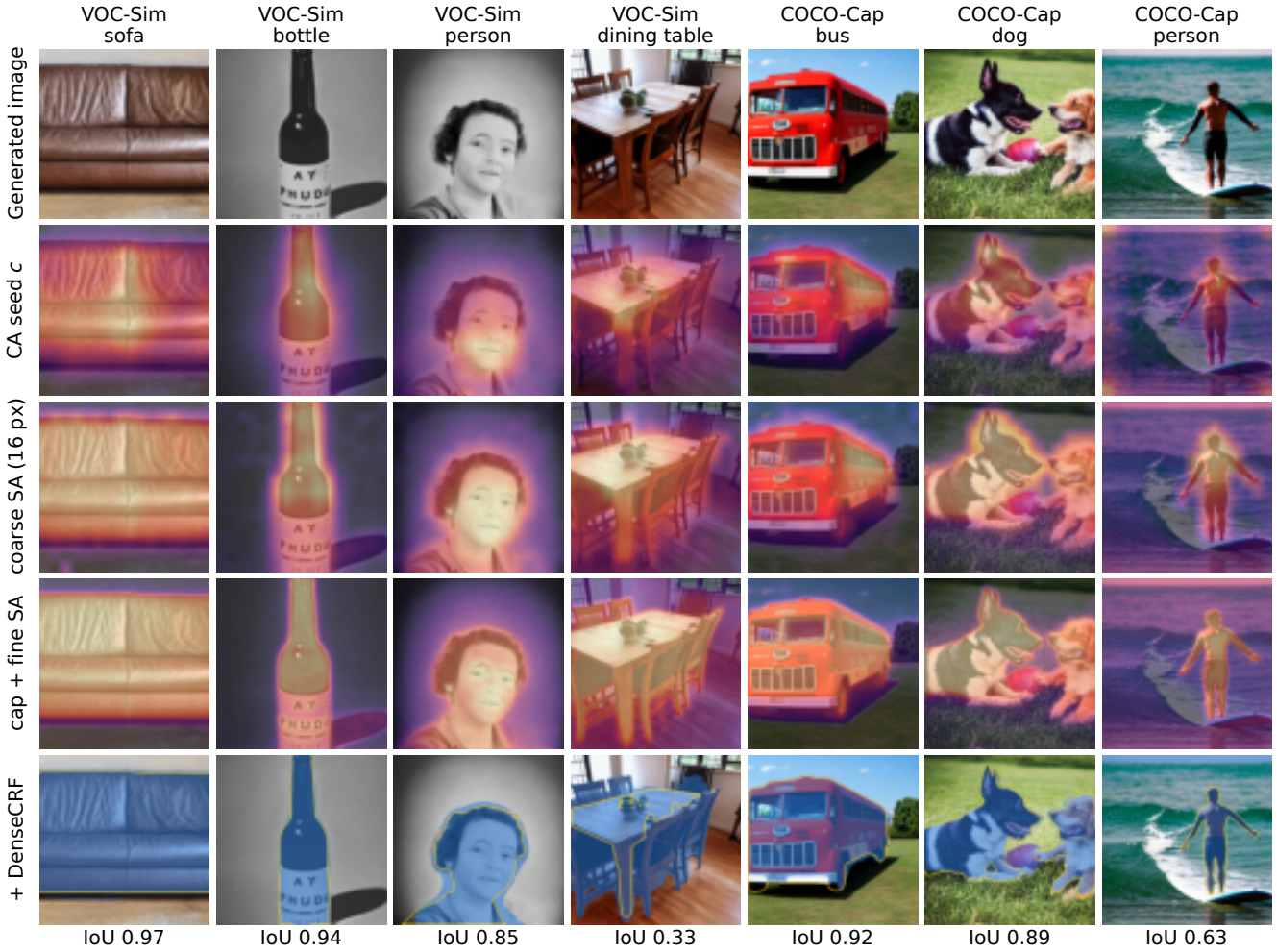


Figure 2. **From seed to mask.** Columns are chosen by a fixed rule, without inspecting any image: the VOC-Sim classes at the best, upper-third, lower-third and worst class IoU of the final pipeline, and the COCO-Cap classes at the best, median and worst, each shown by its median-IoU image (for the worst COCO-Cap class, person, its upper-quartile image). Rows: generated image; A3 CA seed c; after coarse 16-pixel SA (r); after the cap and fine 64-pixel SA (y); final mask after DenseCRF (blue) with the annotation outline (yellow). The VOC dining table shows the main failure mode: SA groups the table with its chairs.

Fig. 3 (right) shows, that mass is almost all background: 13% of the background mass but 0.5% of the object mass, and at most 1.6% for any class. Conservation instead makes children inherit the budget of every parent, including parents that the blurry coarse map wrongly made confident, and it must spend that budget somewhere. Figure 4 shows the effect. Conservation distorts the mask wherever a parent's budget is wrong (COCO cat: IoU 0.58, against 0.81 with the cap), whereas the cap removes mass only where the coarse level disagrees.

5.4. Boundary: what DenseCRF adds

SA works on a 64-pixel grid, so even a good map has blocky, slightly dilated borders (Fig. 5). DenseCRF uses the image itself: the pairwise kernel of Eq. (5) prefers equal labels for nearby pixels of similar colour, so mask boundaries snap to edges. With fixed parameters it improves 278 VOC-Sim images and worsens 97, and raises mIoU by 0.013 (Table 4). Its effect depends on what it is given. On fine-only maps it mainly adds recall and barely helps COCO-Cap (+0.002, with 243 images worse). On capped coarse-to-fine maps it raises precision, helps COCO-Cap six times more (+0.014, 18 of 20 classes), and turns 427 images better against 105 worse. Once the cap has removed far-away mass, the remaining errors lie near boundaries, where appearance is informative. DenseCRF fails when appearance misleads: a chair's thin legs against a similar background (Fig. 5, bottom).

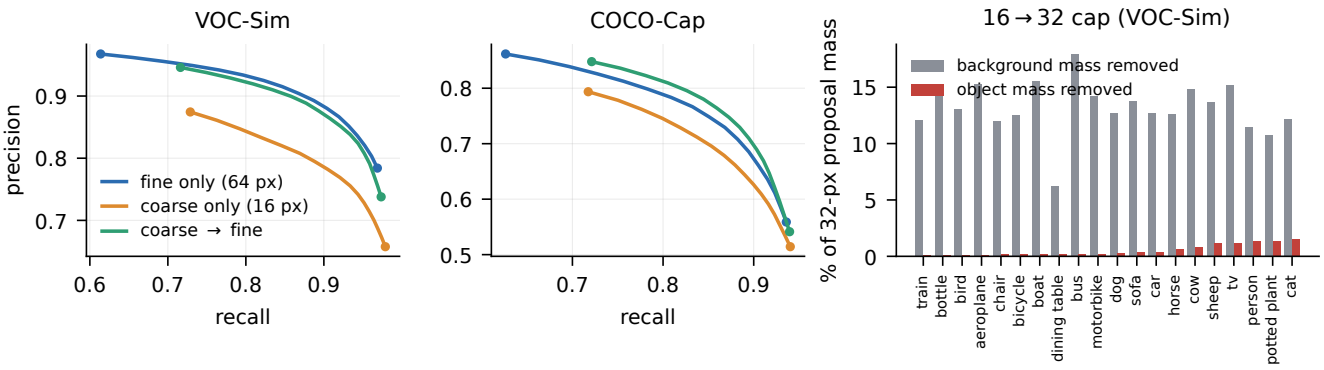


Figure 3. **Coarse SA gives the envelope; the cap trims only background.** *Left, middle:* pooled precision and recall over the threshold grid (0.30 at the high-recall end to 0.70) for one seed propagated by 64-pixel SA only, 16-pixel SA only, or 16→64 pixels (two passes per level, same run). Coarse SA keeps recall at high thresholds and coarse-to-fine combines it with fine precision. *Right:* share of the 32-pixel proposal's mass that the cap removes inside the annotated object (red) and in the background (grey), per VOC-Sim class (ground truth is used only for this diagnostic). Overall 13% of background mass and 0.5% of object mass are removed.

Variant	C2F	conserve (=) 16→32	conserve (=) 32→64	cap (≤) 16→32	cap (≤) 32→64	compl.	wins	mIoU VOC raw	mIoU VOC + R2	mIoU COCO + R2
References										
fine only (64 px)							–	.8098	.8231	.6622
plain hierarchy	✓						1/5	.8061	.8184	.6925
Two-sided: children must carry exactly the parent's mass										
conservation	✓	✓					2/5	.8034	.8185	.7038
conservation	✓		✓				0/5	.7894	.8135	.7022
conservation	✓	✓	✓				0/5	.7592	.7917	.6851
conservation + completion	✓	✓	✓			✓	2/5	.8037	.8204	**.7063**
One-sided: children may not exceed the parent's mass										
cap (ours)	✓			✓			**5/5**	**.8157**	**.8304**	.6966
cap	✓			✓	✓		4/5	.8151	.8290	.6964
cap + completion	✓			✓		✓	3/5	.8131	.8209	.6647

Table 3. **How the coarse level should constrain the finer one.** All rows share one generation of the 490 VOC-Sim and 707 COCO-Cap images, the A3 seed and SA calls 10–20. *C2F*: coarse-to-fine propagation $16 \to 32 \to 64$ px (otherwise 64-px SA only). *Conserve* forces each 2×2 child group to carry exactly its parent's mass (closed-form KL projection). *Cap* only removes mass above the parent (Eq. 4). *Compl.*: fine-SA completion re-adds positive residuals. VOC: forced OOF mIoU; *wins*: held-out folds in which a row beats fine-only after R2; COCO-Cap: VOC-frozen threshold. Forcing parent mass onto children loses up to 5 VOC points. Capping at $16 \to 32$ is the only variant that beats both references in every fold (vs. plain hierarchy: $+0.012$ after R2, paired 95% CI $[+0.006, +0.013]$).

5.5. What the hierarchy changes: calibration across images

The cap barely changes the *ranking* of pixels within an image: raw average precision on COCO-Cap is unchanged (-0.0006). If each image could choose its own threshold (an oracle that uses ground truth), fine-only and coarse-to-fine maps would tie (0.896 vs. 0.895 after R2, Fig. 6). The gain lies in *calibration*: after the cap, one global threshold fits many more images. Removing background mass makes small objects stand out, so the best global threshold moves from 0.425 to 0.55, and at that threshold the curve peaks higher on both benchmarks.

This also explains the one class that loses. Many sheep images are close-ups that fill 80–95% of the frame. Sheep prefer a low threshold (0.35, the best for fine-only maps). At that threshold the capped map leaks into the surroundings, with twice the false positives of the fine-only map. At the shared threshold of 0.55, sheep IoU falls by 0.09. At a matched threshold, the capped map is actually better on sheep (0.807 vs. 0.778 at 0.55), and the cap removes only 1.1% of sheep pixels. We tested six label-free per-image thresholding rules, including Otsu, two-Gaussian fits and area matching. None improved the capped pipeline, and the one that repaired sheep (matching the fine-only map's area) gave back most of the gain (supplementary material). The loss is a calibration trade-off of a single global threshold, not a failure of the propagation.

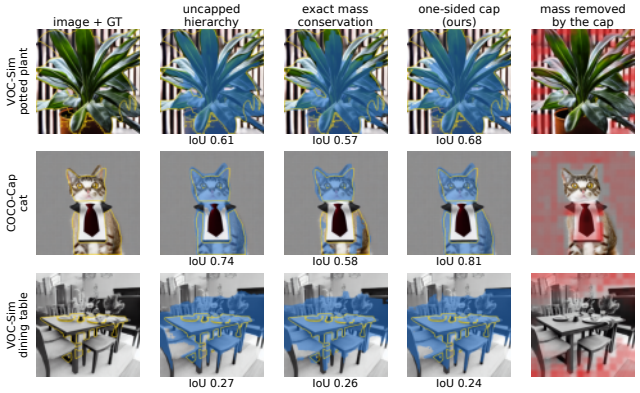


Figure 4. **Cap versus conservation.** Images at the 95th percentile of the cap's IoU gain over the plain hierarchy on VOC-Sim and COCO-Cap, and at the 5th percentile on VOC-Sim (fixed rule). Masks at each variant's VOC-selected threshold. Right: mass removed by the cap (red). The cap removes background mass that the coarse level does not support; conservation spreads mass into it. When the coarse level groups the table with its chairs (bottom), no variant can separate them.

		VOC-Sim			COCO-Cap		
C2F+cap	R2	mIoU	P / R	img ↑/↓	mIoU	P / R	img ↑/↓
		.811	.880 / .904		.660	.708 / .882	
	✓	.824	.883 / .916	291/132	.662	.703 / .889	354/243
✓		.817	.875 / .915		.682	.731 / .889	
✓	✓	**.830**	**.890** / .910	278/97	**.696**	**.747** / .886	427/105

Table 4. **What DenseCRF adds, and why it pairs with the cap.** Full-VOC mIoU, pooled precision/recall and the number of images whose IoU rises/falls by more than 0.005 when R2 is added (same run; thresholds from VOC-Sim, frozen for COCO-Cap). On fine-only maps R2 mainly recovers recall and barely helps COCO-Cap (+0.002). On capped coarse-to-fine maps it raises precision and helps COCO-Cap six times more (+0.014, 18 of 20 classes): with far-away mass removed, the remaining errors lie near object boundaries, where appearance decides.

6. Comparison with published methods

Table 5 places the pipeline beside the class-word results reported by OVAM on the same benchmarks. With class words only and no training, it exceeds OVAM by 0.125 on VOC-Sim and 0.114 on COCO-Cap. It also exceeds DatasetDM, which trains an extra decoder, on both benchmarks. Most of the COCO-Cap margin over our own fine-only pipeline (0.662→0.696) comes from the capped hierarchy, consistent with Sec. 5.2: cluttered scenes benefit most from a coarse envelope. Because we regenerate the images, the published rows are a contextual comparison rather than a paired one; scores of the released baseline masks under our evaluation are in the supplementary material.

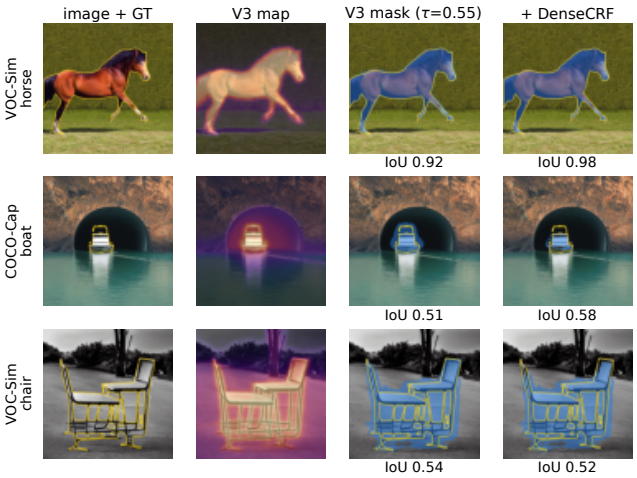


Figure 5. **DenseCRF snaps the mask to image edges.** Images at the 90th percentile of the DenseCRF gain on VOC-Sim and COCO-Cap, and at the 10th percentile on VOC-Sim (fixed rule).

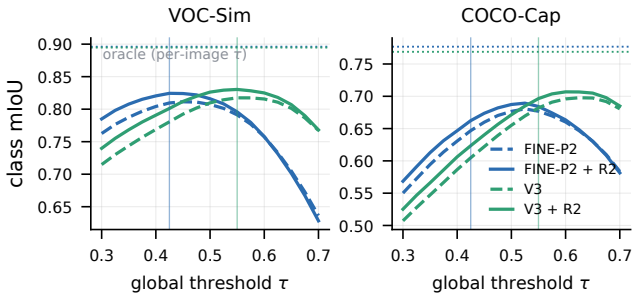


Figure 6. **The hierarchy improves calibration, not ranking.** Class mIoU against the global threshold for fine-only (FINE-P2) and coarse-to-fine (V3) maps, raw (dashed) and with DenseCRF (solid), on all images of one run. Vertical lines: VOC-selected thresholds (0.425 and 0.55). Dotted: per-image best threshold (oracle, uses ground truth), which ties the two maps.

7. Discussion and limitations

What the negative results taught us. Before arriving at the hierarchy, we registered and ran 45 comparisons of alternative operators, most of them on top of fine-only propagation (supplementary material). These included absorbing random walks, confidence-locked or reciprocal propagation, SA region labelling, graph sharpening, observation-reliability weighting, stopped random walks, re-seeding between scales and Markov arrival times. Several improved the raw map, but apart from the SA call window none survived DenseCRF: refinement absorbed gains that only completed objects. This is why the present pipeline is simple, and why we select on the *refined* mask. The capped hierarchy is the largest change that survived refinement in every fold and also transferred.

Method	VOC-Sim	COCO-Cap
Training-free, class word (published in OVAM, Table 1)		
DAAM [10]	.662	.484
Attn2Mask	.687	.550
OVAM [6]	.704	.582
With extra training (published in OVAM, Table 1)		
Grounded Diffusion	.621	.502
DatasetDM	.803	.593
Ours, training-free, class word		
fine-only propagation + R2	.823	.662
coarse-to-fine + cap + R2	**.829**	**.696**

Table 5. **Comparison with published class-word results** on the two benchmarks released by OVAM [6] (same prompts, seeds and annotations; optimized-token rows are excluded because they use mask supervision). Our VOC-Sim values are forced out-of-fold estimates; COCO-Cap is frozen transfer. Because our method records attention during generation, we regenerate each image from its released prompt and seed; renders can differ slightly from those annotated, so cross-block differences are indicative rather than paired.

Failure modes. The pipeline inherits SA's notion of an object. When co-occurring objects share coarse affinity (the dining table and its chairs, Figs. 2 and 4) or the seed is weak (COCO-Cap person, our weakest class at 0.28 IoU), propagation fills the wrong region. Frame-filling objects lose under one global threshold (Sec. 5.5).

Scope. Both benchmarks contain generated images, regenerated from released prompts and seeds; renders can differ slightly from the annotated ones. All choices are made on VOC-Sim with one backbone, Stable Diffusion 1.5. Two components were adopted after the registered rule kept a simpler reference (Sec. 4). We report forced OOF estimates for them and the registered outcomes alongside. Their gains hold in 4/5 and 5/5 held-out folds and transfer to COCO-Cap, but they are not pre-registered promotions.

8. Conclusion

Training-free segmentation from diffusion attention works best when each scale does what it is good at. Coarse CA and SA decide *where* the object is; finer SA decides *exactly where* its boundary lies, but only within the mass that the coarse level allows; and DenseCRF aligns that boundary with the image. The one-sided cap is the piece that makes this division of labour work, and it improves cross-image calibration, the main bottleneck of a single global threshold.

References

[1] Huy Che and Vinh-Tiep Nguyen. FA-Seg: A fast and accurate diffusion-based method for open-vocabulary segmentation, 2025. 2

[2] Markus Karmann and Onay Urfalioglu. Repurposing stable diffusion attention for training-free unsupervised interactive segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 2

[3] Philipp Krähenbühl and Vladlen Koltun. Efficient inference in fully connected crfs with gaussian edge potentials. In *Advances in Neural Information Processing Systems*, 2011. 2, 3

[4] Kunal Mahatha, Jose Dolz, and Christian Desrosiers. SPARK: Stochastic propagation via affinity-guided random walk for training-free unsupervised segmentation, 2026. 1, 2

[5] Kunal Mahatha B, Jose Dolz, and Christian Desrosiers. NERVE: Neighbourhood and entropy-guided random-walk for training free open-vocabulary segmentation, 2025. 1, 2

[6] Pablo Marcos-Manchón, Roberto Alcover-Couso, Juan C. SanMiguel, and José M. Martínez. Open-vocabulary attention maps with token optimization for semantic segmentation in diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9242–9252, 2024. 1, 2, 3, 8

[7] Benyuan Meng, Qianqian Xu, Zitai Wang, Xiaochun Cao, Longtao Huang, and Qingming Huang. Making training-free diffusion segmentors scale with the generative power. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 35861–35871, 2026. 1, 2

[8] Joon Hyun Park, Kumju Jo, and Sungyong Baik. SeeDiff: Off-the-shelf seeded mask generation from diffusion models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 6406–6415, 2025. 1, 2

[9] Gensheng Pei, Xiruo Jiang, Xinhao Cai, Tao Chen, Yazhou Yao, and Byeungwoo Jeon. PEARL: Geometry aligns semantics for training-free open-vocabulary semantic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17927–17937, 2026. 2

[10] Raphael Tang, Linqing Liu, Akshat Pandey, Zhiying Jiang, Gefei Yang, Karun Kumar, Pontus Stenetorp, Jimmy Lin, and Ferhan Ture. What the daam: Interpreting stable diffusion using cross attention. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 5644–5659, 2023. 1, 2, 8